

# Marcos Esparza

432-661-7620 | [esparza\\_m58311@utpb.edu](mailto:esparza_m58311@utpb.edu) | [LinkedIn](#) | [Portfolio](#) | U.S. Citizen

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## Education

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### Bachelor of Science: Mechanical Engineering

Aug 2022 - December 2026

University of Texas Permian Basin | GPA: 3.0

Odessa, TX

Relevant Coursework: Thermodynamics II, Fluid Mechanics II, Mechanics of Materials, Thermo-Fluids Lab, Engineering Design

Activities: FAST, ASME, AIAA, SPE

## Technical Projects

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### Falcon Aeronautics & Space Team (FAST) – IREC 10k COTS Rocket

Aerodynamic Design Lead – Fin Can & Structures | UTPB | Aug 2024 – Present

- Leading aerodynamic and fin-can design for FAST's 10,000-ft COTS rocket (Spaceport America Cup & Lone Star Cup), including fin geometry, stability, and airframe layout in SOLIDWORKS/OpenRocket, and coordinating design decisions with propulsion and recovery subteams.
- Performing MATLAB-based fin-flutter and stability analysis, maintaining  $\sim 2.8$ -caliber static margin and  $\geq 1.5\times$  fin-flutter safety factor at peak flight velocity.

### Thermo-Fluids / Mechanical Systems Lab – UTPB

Undergraduate Lab Engineer | Aug 2025 – Dec 2025

- Ran experiments on Pelton turbine, cross-flow heat exchanger (TE93), flow-meter rigs, and fluid-friction/bend-loss apparatus (H408) to measure efficiency, head loss, and heat-transfer or flow coefficients.
- Processed pressure, temperature, and flow data in MATLAB and Excel to compare results with theory, quantify error, and document system performance and uncertainty using standard correlations (Bernoulli, Moody/Blasius, Nusselt).
- Analyzed sample data from a Turbine Technologies SR-30 turbojet (Mini-Lab) for a Brayton-cycle study, estimating inlet/exit Mach number, mass flow rate, and theoretical thrust.

### L2 High-Power Rocket (In-Progress Personal Project)

- Designing a 4" dual-deploy rocket with fiberglass-reinforced cardboard airframe, 38 mm motor mount, and TTW fins for an upcoming NAR/Tripoli Level 2 certification flight.
- Modeling performance in OpenRocket ( $\approx 4,200$  ft apogee on J425) and sizing a 36" main and 14" drogue to achieve safe descent rates and meet Level 2 certification requirements.

## Experience

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### Logistics Associate — Academy Sports + Outdoors, Midland, TX

Jul 2024 – Present

- Process and stage online and in-store pickup orders, consistently meeting same-day deadlines.
- Perform cycle counts and resolve stock discrepancies to maintain accurate inventory.

### Overnight Deli Production — H-E-B, Midland, TX

Oct 2022 – Jan 2024

- Prepared and stocked deli products overnight to support next-day store operations while helping train new team members on shift procedures and safe handling practices.

### Team Member — Whataburger, Midland, TX

Oct 2020 – Mar 2022

- Supported front counter and kitchen operations during high-volume shifts while maintaining order accuracy and cleanliness.

## Technical Skills

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**Programming:** MATLAB, Python (basic), LabVIEW (exposure)

**Software:** SOLIDWORKS, OpenRocket, Excel, Word, PowerPoint

**Test & Lab:** Pressure/temperature instrumentation, flow meters (orifice, Venturi, rotameter), Pelton turbine rig, cross-flow heat exchanger (TE93), fluid-friction/bend-loss apparatus (H408), SR-30 turbojet data analysis

**Fabrication:** Basic machining & hand tools, 3D printing, epoxy layups, high-power rocketry construction (TTW fins, dual-deploy, avionics integration)

**Engineering:** Aerodynamic & structural design, test data analysis, systems integration, mechanical assembly, technical writing

**Certifications:** Tripoli/NAR High-Power Rocketry Level 1; Level 2 in progress